FSC CLASS 5305

REVIEW: ME, EA, IS, AV, MI, AT
USER: SH, AS, GL, MU, YD, PA, WY, MC

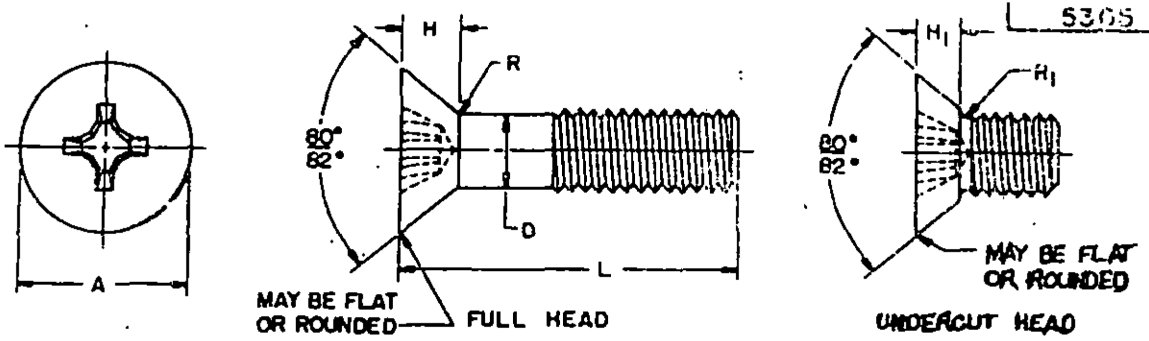

TABLE I

NOMINAL SIZE		(#0) .060	(#2) .086	(#4) .112	(#6) .138	(#8) .164	(#10) .190	(1/4) .250	(5/16) .3125	(3/8) .375
THREADS PER INCH, UNF-2A		80	64	48	40	36	32	28	24	24
D BODY DIAMETER	MAX MIN	.0600 .0496	.0860 .0733	.1120 .0954	.1380 .1184	.1640 .1424	.1900 .1658	.2500 .2225	.3125 .2806	.3750 .3430
A HEAD DIAMETER	MAX SHARP MIN SHARP ABS MIN	.119 .105 .099	.172 .156 .147	.225 .207 .195	.279 .257 .244	.332 .308 .292	.385 .359 .340	.507 .477 .452	.635 .600 .568	.762 .722 .685
H HEAD HEIGHT	REF	.035	.051	.067	.083	.100	.116	.153	.191	.230
H_1 HEAD HEIGHT UNDERCUT	MAX MIN	.025 .018	.036 .028	.047 .038	.059 .048	.070 .058	.081 .068	.107 .092	.134 .116	.161 .140
R RADIUS	MAX	.024	.034	.045	.055	.066	.076	.100	.125	.150
R_1 RADIUS-UNDERCUT	MAX	.009	.013	.017	.021	.025	.029	.038	.047	.056
DRIVER SIZE		0	1	1	2	2	2	3	4	4
TENSILE STRENGTH (LBS)	MIN	100	220	380	580	840	1140	2070	3510	5000
L LENGTH	TOLERANCE	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
.125	+0 -.031	109	1	10	21	36				
.188		110	2	11	22	37				
.250		111	3	12	23	38	51			
.312		112	4	13	24	39	52	66		
.375		113	5	14	25	40	53	67	80	
.438			6	15	26	41	54	68	81	
.500			7	16	27	42	55	69	82	93
.625			8	17	28	43	56	70	83	94
.750			9	18	29	44	57	71	84	95
.875				19	30	45	58	72	85	96
1.000				20	31	46	59	73	86	97
1.250	+0 -.062				32	47	60	74	87	98
1.500					33	48	61	75	88	99
1.750					34	49	62	76	89	100
2.000					35	50	63	77	90	101
2.250	+0 -.094						64	78	91	102
2.500							65	79	92	103
2.750										104
3.000										105

NOTES:

1. MATERIAL: Brass in accordance with QQ-B-626 or QQ-W-321 having minimum ultimate tensile strength of 57,000 PSI.
2. PROTECTIVE COATING: Black Chemical Finish in accordance with MIL-F-495.
3. THREADS: The threads shall be in accordance with Screw-Thread Standards for Federal Services, Handbook H-28.
4. THREAD LENGTH. Screws 2.00 inches long and shorter shall have complete threads extending to within two threads of the head, or closer if practicable. Longer screws shall have a minimum thread length of 1.50 inches.
5. RECESS: The recess shall be in accordance with MS9006.
6. DIMENSIONS: All dimensions are in inches unless otherwise specified.
7. UNDERCUT HEADS: Screws above dotted lines shall have undercut heads.
8. PART NUMBER: The MS part number consists of the MS number, plus the dash number. Example: MS35199-1.
9. Referenced documents shall be of the issue in effect on date of invitations for bid.
10. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

Ⓑ ENTIRE STANDARD REVISED

APPROVED 25 JUN 53 REVISED Ⓐ 21 MAR 63 Ⓑ 14 JUN 1976

Military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document, when applicable.

P.A. WC
Other Cust
82

TITLE
SCREW, MACHINE-FLAT COUNTERSUNK HEAD, 82°, CROSS-RECESSED, BRASS, BLACK CHEMICAL FINISH, UNF-2A

MILITARY STANDARD
MS35199

PROCUREMENT SPECIFICATION
FF-S-92

SUPERSEDES: [illegible] IN PART, [illegible] IN PART, [illegible] IN PART.

SHEET 1 OF 4

DD FORM 672-1 (COORDINATED)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

5305-1212

FED. SUP CLASS
5305

TABLE II

The screws covered by the dash numbers given in MS35197, MS35246, MS35248 and AN510 in part are cancelled/inactivated after the dates indicated on the documents. Use the dash numbers given in the preceding sheets. The cancelled screws cannot always replace the superseding screws and should be used until existing stocks are depleted. Use only the superseding screws for design and replacement. Replacement shall be in accordance with this table.

CROSS REFERENCE OF PART NUMBERS

CANCELLED				SUPERSEDED BY
MS35197	MS35246 MS35248	AN510		MS35199
Dash Number				Dash Number
	1	B0R2	B0-2	109
	2	B0R3	B0-3	110
	3	B0R4	B0-4	111
	4	B0R5	B0-5	112
	5	B0R6	B0-6	113
		B0R7	B0-7	No Replacement
		B0R8	B0-8	No Replacement
	6	B1R2	B1-2	No Replacement
	7	B1R3	B1-3	No Replacement
	8	B1R4	B1-4	No Replacement
	9	B1R5	B1-5	No Replacement
	10	B1R6	B1-6	No Replacement
	11	B1R7	B1-7	No Replacement
	12	B1R8	B1-8	No Replacement
1	13			1
2	14	B2R3	B2-3	2
3	15	B2R4	B2-4	3
	16	B2R5	B2-5	4
5	17	B2R6	B2-6	5
6	18	B2R7	B2-7	6
7	19	B2R8	B2-8	7
		B2R9	B2-9	No Replacement
8	20	B2R10	B2-10	5
		B2R11	B2-11	No Replacement
9	21	B2R12	B2-12	9
		B3R3	B3-3	No Replacement
		B3R4	B3-4	No Replacement
		B3R5	B3-5	No Replacement
		B3R6	B3-6	No Replacement
		B3R7	B3-7	No Replacement
		B3R8	B3-8	No Replacement
		B3R9	B3-9	No Replacement
		B3R10	B3-10	No Replacement
		B3R11	B3-11	No Replacement
		B3R12	B3-12	No Replacement
10	22			10
11	23	B4R3	B4-3	11
12	24	B4R4	B4-4	12
13	25	B4R5	B4-5	13
14	26	B4R6	B4-6	14
15	27	B4R7	B4-7	15
16	28	B4R8	B4-8	16
		B4R9	B4-9	No Replacement
17	29	B4R10	B4-10	17
		B4R11	B4-11	No Replacement
18	30	B4R12	B4-12	18
		B4R13	B4-13	No Replacement
19	31	B4R14	B4-14	19
		B4R15	B4-15	No Replacement
20	32	B4R16	B4-16	20
		B4R18	B4-18	No Replacement
		B4R20	B4-20	No Replacement
		B4R22	B4-22	No Replacement
		B4R24	B4-24	No Replacement
		B4R26	B4-26	No Replacement
		B4R28	B4-28	No Replacement
		B5R4	B5-4	No Replacement
		B5R5	B5-5	No Replacement
		B5R6	B5-6	No Replacement
		B5R7	B5-7	No Replacement
		B5R8	B5-8	No Replacement
		B5R9	B5-9	No Replacement
		B5R10	B5-10	No Replacement
		B5R11	B5-11	No Replacement
		B5R12	B5-12	No Replacement
		B5R13	B5-13	No Replacement
		B5R14	B5-14	No Replacement
		B5R15	B5-15	No Replacement
		B5R16	B5-16	No Replacement
		B5R18	B5-18	No Replacement
		B5R20	B5-20	No Replacement
		B5R22	B5-22	No Replacement
		B5R24	B5-24	No Replacement
		B5R26	B5-26	No Replacement
		B5R28	B5-28	No Replacement
21	33			21
22	34			22
23	35	B6R4	B6-4	23
24	36	B6R5	B6-5	24
25	37	B6R6	B6-6	25
26	38	B6R7	B6-7	26
27	39	B6R8	B6-8	27
		B6R9	B6-9	No Replacement
28	40	B6R10	B6-10	28
		B6R11	B6-11	No Replacement
29	41	B6R12	B6-12	29
		B6R13	B6-13	No Replacement
30	42	B6R14	B6-14	30
		B6R15	B6-15	No Replacement
31	43	B6R16	B6-16	31
		B6R18	B6-18	No Replacement
32	44	B6R20	B6-20	32
		B6R22	B6-22	No Replacement
33	45	B6R24	B6-24	33
		B6R26	B6-26	No Replacement
34	46	B6R28	B6-28	34
		B6R30	B6-30	No Replacement
35	47	B6R32	B6-32	35
		B6R34	B6-34	No Replacement
		B6R36	B6-36	No Replacement
		B6R38	B6-38	No Replacement
		B6R40	B6-40	No Replacement
36	48			36
37	49			37
38	50	B8R4	B8-4	38
39	51	B8R5	B8-5	39
40	52	B8R6	B8-6	40
41	53	B8R7	B8-7	41
42	54	B8R8	B8-8	42
		B8R9	B8-9	No Replacement
43	55	B8R10	B8-10	43
		B8R11	B8-11	No Replacement

(Continued on Sheet 3)

APPROVED 25 JUNE 53 REVISED (A) 21 MAR 1963 (B) FOR CHANGES SEE SHEET 1.

REVIEW: ME, EA, SH, AV, MI, AT
USER: SH, AR, OL, MU, YD, PA, WY, MC

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document, [illegible].

P.A. Other Cust	WC 82	TITLE SCREW, MACHINE-FLAT COUNTERSUNK HEAD, 82°, CROSS-RECESSED, BRASS, BLACK CHEMICAL FINISH, UNF-2A
PROCUREMENT SPECIFICATION FF-S-92		SUPERSEDES: MS35197, MS35246, MS35248, AN510 IN PART, [illegible] IN PART, [illegible]

MILITARY STANDARD

MS35199

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

FED. SUP CLASS
5305

CROSS REFERENCE OF PART NUMBERS

CANCELLED				SUPERSEDED BY
MS35197	MS35246 MS35248	AN510		MS35199
Dash Number				Dash Number
44	56	B8R12 B8R13	B8-12 B8-13	44 No Replacement
45	57	B8R14 B8R15	B8 14 B8-15	45 No Replacement
46	58	B8R16 B8R18	B8-16 B8-18	46 No Replacement
47	59	B8R20 B8R22	B8-20 B8-22	47 No Replacement
48	60	B8R24 B8R26	B8-24 B8-26	48 No Replacement
49	61	B8R28 B8R30	B8-28 B8-30	49 No Replacement
50	62	B8R32 B8R34	B8-32 B8-34	50 No Replacement
	63	B8R36 B8R38	B8-36 B8-38	No Replacement No Replacement
	64	B8R40 B8R42	B8-40 B8-42	No Replacement No Replacement
	65	B8R44 B8R46	B8-44 B8-46	No Replacement No Replacement
	66	B8R48	B8-48	No Replacement
	67			No Replacement
51	68	B10R4	B10-4	51
52	69	B10R5	B10-5	52
53	70	B10R6	B10-6	53
54	71	B10R7	B10-7	54
55	72	B10R8 B10R9	B10-8 B10-9	55 No Replacement
56	73	B10R10 B10R11	B10-10 B10-11	56 No Replacement
57	74	B10R12 B10R13	B10-12 B10-13	57 No Replacement
58	75	B10R14 B10R15	B10-14 B10-15	58 No Replacement
59	76	B10R16 B10R18	B10-16 B10-18	59 No Replacement
60	77	B10R20 B10R22	B10-20 B10-22	60 No Replacement
61	78	B10R24 B10R26	B10-24 B10-26	61 No Replacement
62	79	B10R28 B10R30	B10-28 B10-30	62 No Replacement
63	80	B10R32 B10R34	B10-32 B10-34	63 No Replacement
64	81	B10R36 B10R38	B10-36 B10-38	64 No Replacement
65	82	B10R40 B10R42	B10-40 B10-42	65 No Replacement
	83	B10R44	B10-44	No Replacement
		B10R46	B10-46	No Replacement
	84	B10R48	B10-48	No Replacement
	85			No Replacement
66	86			66
67	87	B416R6	B416-6	67
68	88	B416R7	B416-7	68
69	89	B416R8	B416-8	69

CANCELLED				SUPERSEDED BY
MS35197	MS35246 MS35248	AN510		MS35199
Dash Number				Dash Number
		B416R9	B416-9	No Replacement
70	90	B416R10 B416R11	B416-10 B416-11	70 No Replacement
71	91	B416R12 B416R13	B416-12 B416-13	71 No Replacement
72	92	B416R14 B416R15	B416-14 B416-15	72 No Replacement
73	93	B416R16 B416R18	B416-16 B416-18	73 No Replacement
74	94	B416R20 B416R22	B416-20 B416-22	74 No Replacement
75	95	B416R24 B416R26	B416-24 B416-26	75 No Replacement
76	96	B416R28 B416R30	B416-28 B416-30	76 No Replacement
77	97	B416R32 B416R34	B416-32 B416-34	77 No Replacement
78	98	B416R36 B416R38	B416-36 B416-38	78 No Replacement
79	99	B416R40 B416R42	B416-40 B416-42	79 No Replacement
	100	B416R44 B416R46	B416-44 B416-46	No Replacement No Replacement
	101	B416R48	B416-48	No Replacement
80	102			80
81	103			81
82	104			82
83	105			83
84	106			84
85	107			85
86	108			86
87	109			87
88	110			88
89	111			89
90	112			90
91	113			91
92	114			92
	115			No Replacement
	116			No Replacement
93	117			93
94	118			94
95	119			95
96	120			96
97	121			97
98	122			98
99	123			99
100	124			100
101	125			101
102	126			102
103	127			103
104	128			104
105	129			105
	130 thru 131			No Replacement

REVIEW: ME, EA, IS, AV, MI, AT
USER: SH, AS, GL, MU, YD, MA, WV, MC

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APPROVED 23 JUNE 53 REVISED Ⓐ 21 MAR 1963 Ⓑ FOR CHANGES SEE SHEET 1.

P.A. WC Other Cust 82	TITLE SCREW, MACHINE-FLAT COUNTERSUNK HEAD, 82°, CROSS-RECESSED, BRASS, BLACK CHEMICAL FINISH, UNF-2A	MILITARY STANDARD **MS35199**
PROCUREMENT SPECIFICATION FF-S-92	SUPERSEDES: MS35197, MS35246, MS35248, MS35368 IN PART, MS35343 IN PART, AN510 [illegible] IN PART, [illegible] IN PART, [illegible] AND [illegible] IN PART.	SHEET 3 OF 4

FED. SUP CLASS
5305

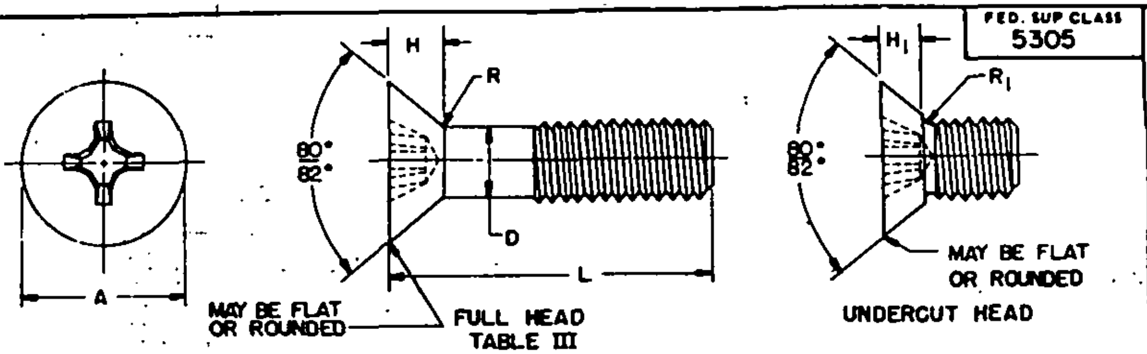

TABLE III
METRIC TRANSLATION OF TABLE I

		1.524	2.184	2.845	3.505	4.166	4.826	6.350	7.938	9.525
NOMINAL SIZE	MM	1.524	2.184	2.845	3.505	4.166	4.826	6.350	7.938	9.525
PITCH		.32	.40	.53	.64	.71	.79	.91	1.06	1.06
D BODY DIAMETER	MAX MIN	1.524 1.260	2.184 1.862	2.845 2.423	3.505 3.007	4.166 3.617	4.826 4.211	6.350 5.652	7.938 7.127	9.525 8.712
A HEAD DIAMETER	MAX SHARP MIN SHARP ABS MIN	3.023 2.667 2.515	4.369 3.962 3.734	5.715 5.258 4.923	7.087 6.528 6.198	8.433 7.823 7.417	9.779 9.119 8.636	12.878 12.116 11.481	16.129 15.240 14.427	19.355 18.339 17.399
H HEAD HEIGHT	REF	.889	1.295	1.702	2.108	2.540	2.946	3.886	4.851	5.842
H1 HEAD HEIGHT UNDERCUT	MAX MIN	.635 .457	.914 .711	1.194 .965	1.499 1.219	1.778 1.473	2.057 1.727	2.718 2.337	3.404 2.946	4.089 3.555
R RADIUS	MAX	.610	.864	1.143	1.397	1.676	1.930	2.540	3.175	3.810
R1 RADIUS-UNDERCUT	MAX	.229	.330	.432	.533	.635	.737	.965	1.194	1.422
DRIVER SIZE		0	1	1	2	2	2	3	4	4
TENSILE STRENGTH, KN-MIN		.448	.986	1.703	2.599	3.738	5.109	9.278	14.836	22.411
L LENGTH	TOLERANCE	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
3.175	+0 -.787	109	1	10	21	36				
4.775		110	2	11	22	37				
6.350		111	3	12	23	38	51			
7.925		112	4	13	24	39	52	66		
9.525		113	5	14	25	40	53	67	80	
11.125			6	15	26	41	54	68	81	
12.700			7	16	27	42	55	69	82	93
15.875			8	17	28	43	56	70	83	94
19.050			9	18	29	44	57	71	84	95
22.225				19	30	45	58	72	85	96
25.400				20	31	46	59	73	86	97
31.750	+0 -1.575				32	47	60	74	87	98
38.100					33	48	61	75	88	99
44.450					34	49	62	76	89	100
50.800					35	50	63	77	90	101
57.150	+0 -2.388						64	78	91	102
63.500							65	79	92	103
69.850										104
76.200										105

NOTE: All dimensions are in millimetres (mm), unless otherwise specified, and are based on an approximate conversion factor of 25.4mm equals one inch.

APPROVED 25 JUN 1953   REVISED Ⓐ 21 MAR 1963 Ⓑ FOR CHANGES SEE SHEET 1.

P.A. WC Other Cust 82	TITLE SCREW, MACHINE-FLAT COUNTERSUNK HEAD, 82°, CROSS-RECESSED, BRASS, BLACK CHEMICAL FINISH, UNF-2A	MILITARY STANDARD MS35199
PROCUREMENT SPECIFICATION FF-S-92	SUPERSEDES: [illegible] IN PART, [illegible] IN PART	SHEET 4 OF 4

REVIEW: ME, EA, IS, AV, MI, AT
USER: SH, AS, OL, MU, YD, PA, WV, MC

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document, [illegible].

DD FORM 672-1 (COORDINATED)